

lab 8

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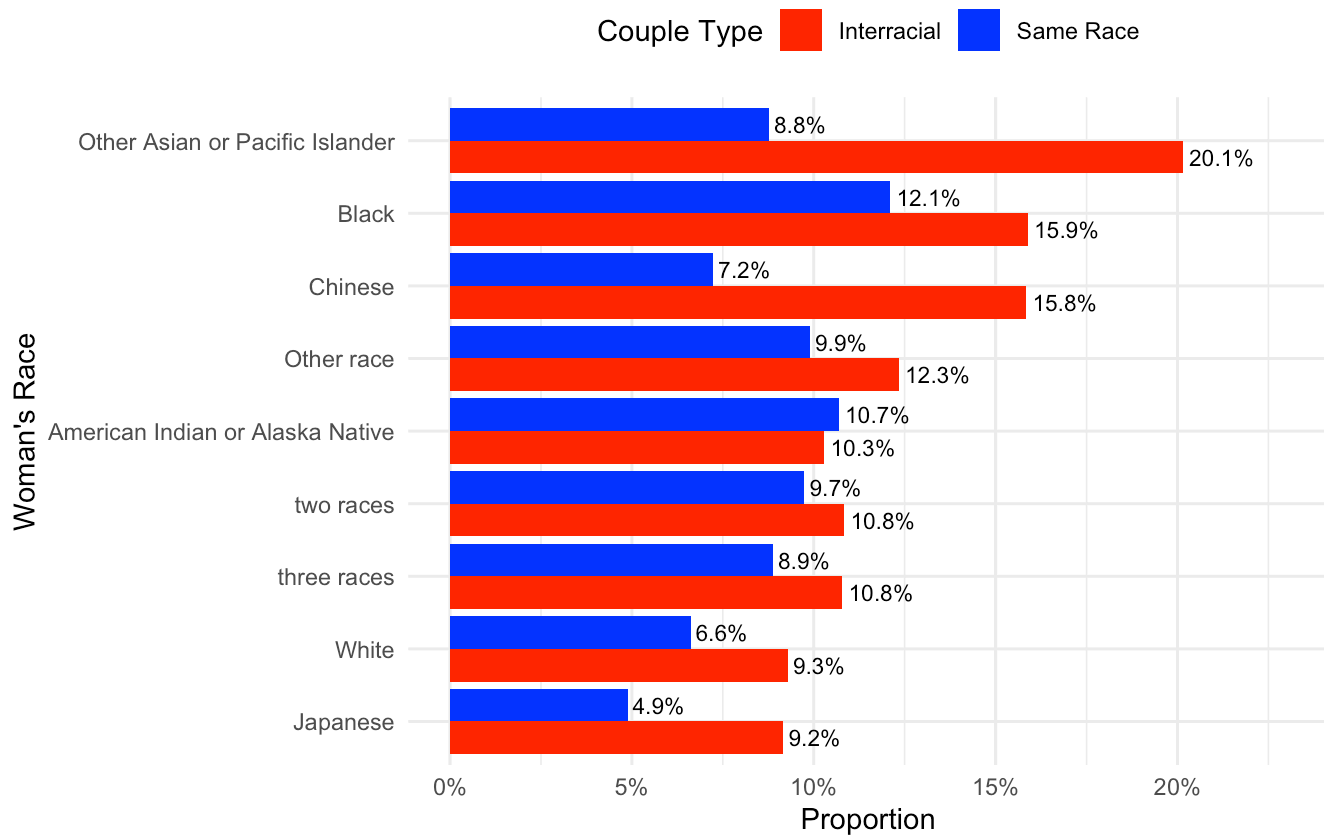
2025-11-14

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ lubridate  1.9.3      ✓ tibble     3.2.1
## ✓ purrr      1.0.2      ✓ tidyr      1.3.1
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
  to become errors
## Type 'citation("pROC")' for a citation.
##
##
## Attaching package: 'pROC'
##
##
## The following objects are masked from 'package:stats':
##
##      cov, smooth, var
```

In this assignment, we attempt to see what parameters we can use to find if a given male household head is 10 years or older than his female partner.

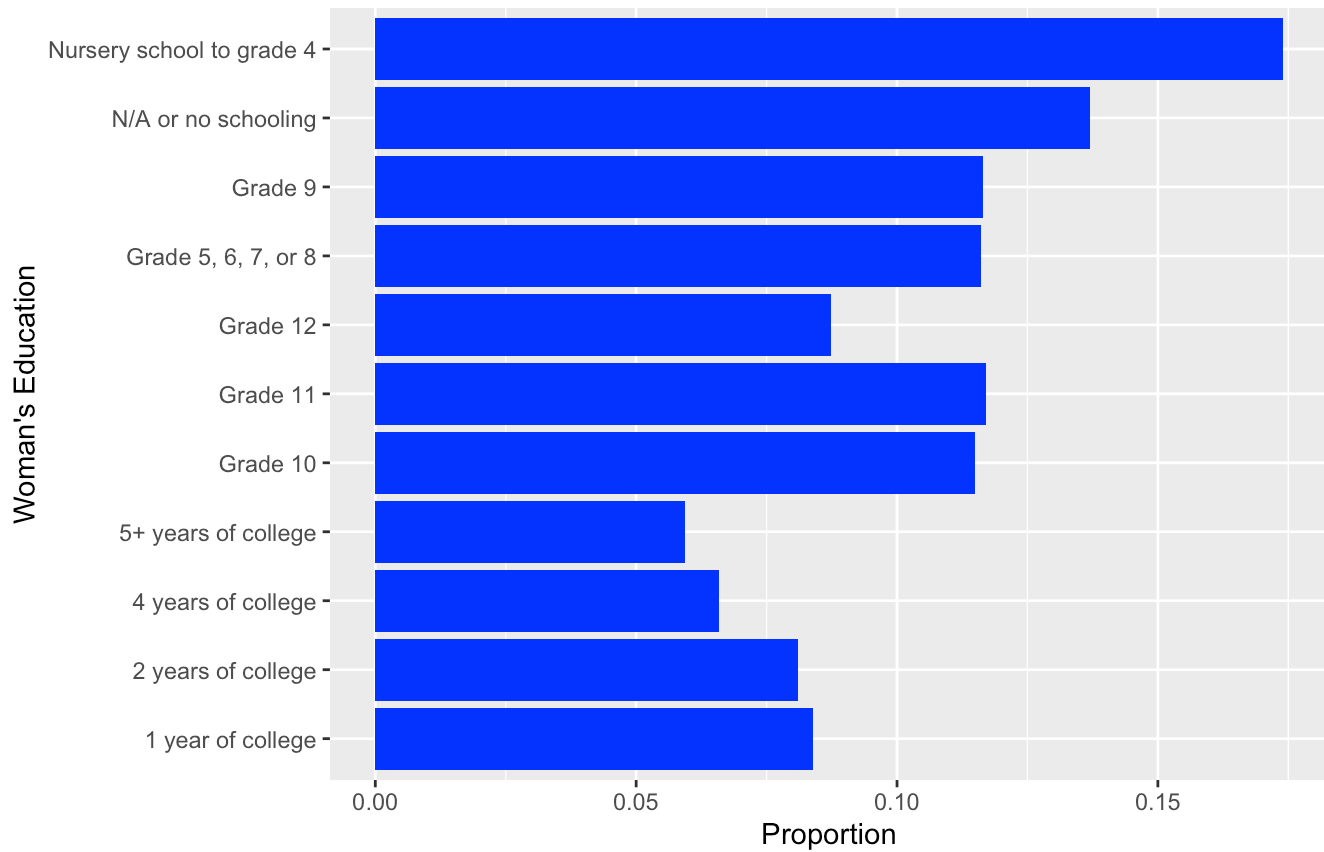
We first examine whether couple type (interracial vs. same-race) and race are associated with the proportion of women married to men more than 10 years older. Examined plots below indicate that certain groups have higher proportion of women married to men who are 10 years or older.

Proportion of Couples Where Man is 10+ Years Older By Woman's Race and Interracial Status

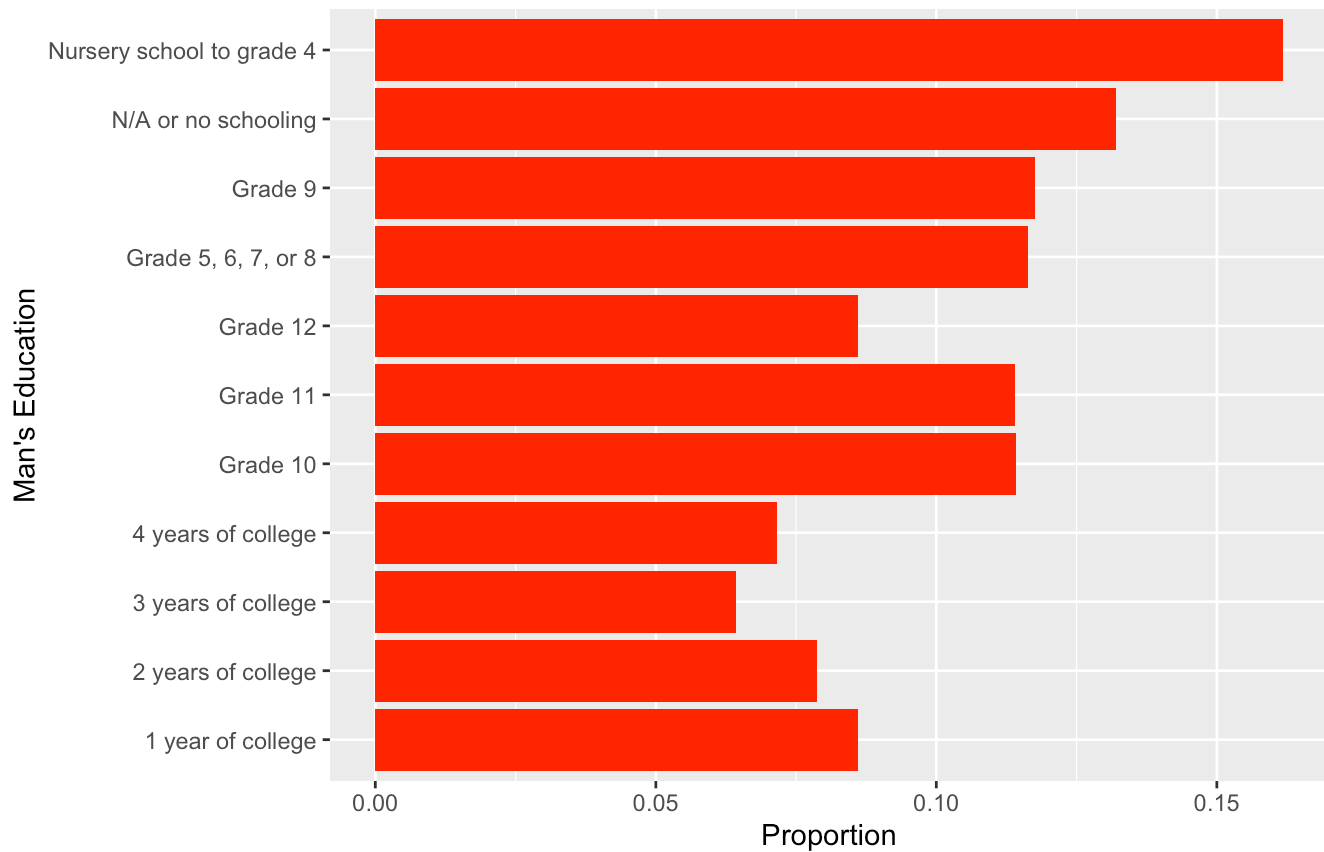


We also see a pattern where higher proportion of women with lower education level tend to marry men who are 10 years or older. However, the proportion is very low as the overall dataset has lower than 1% of the population where a woman is with a man who is 10 years or older.

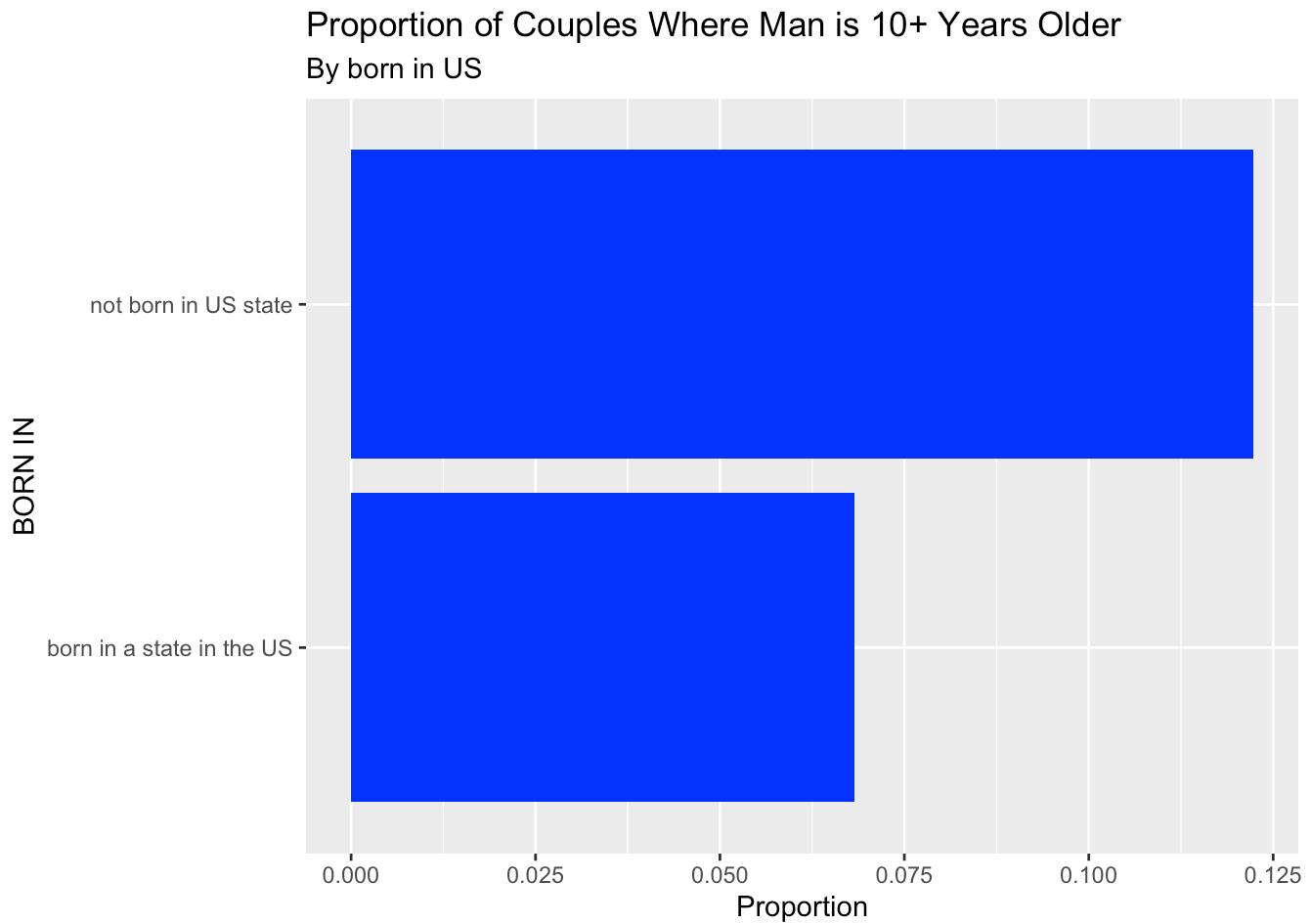
Proportion of Couples Where Man is 10+ Years Older By Woman's Education



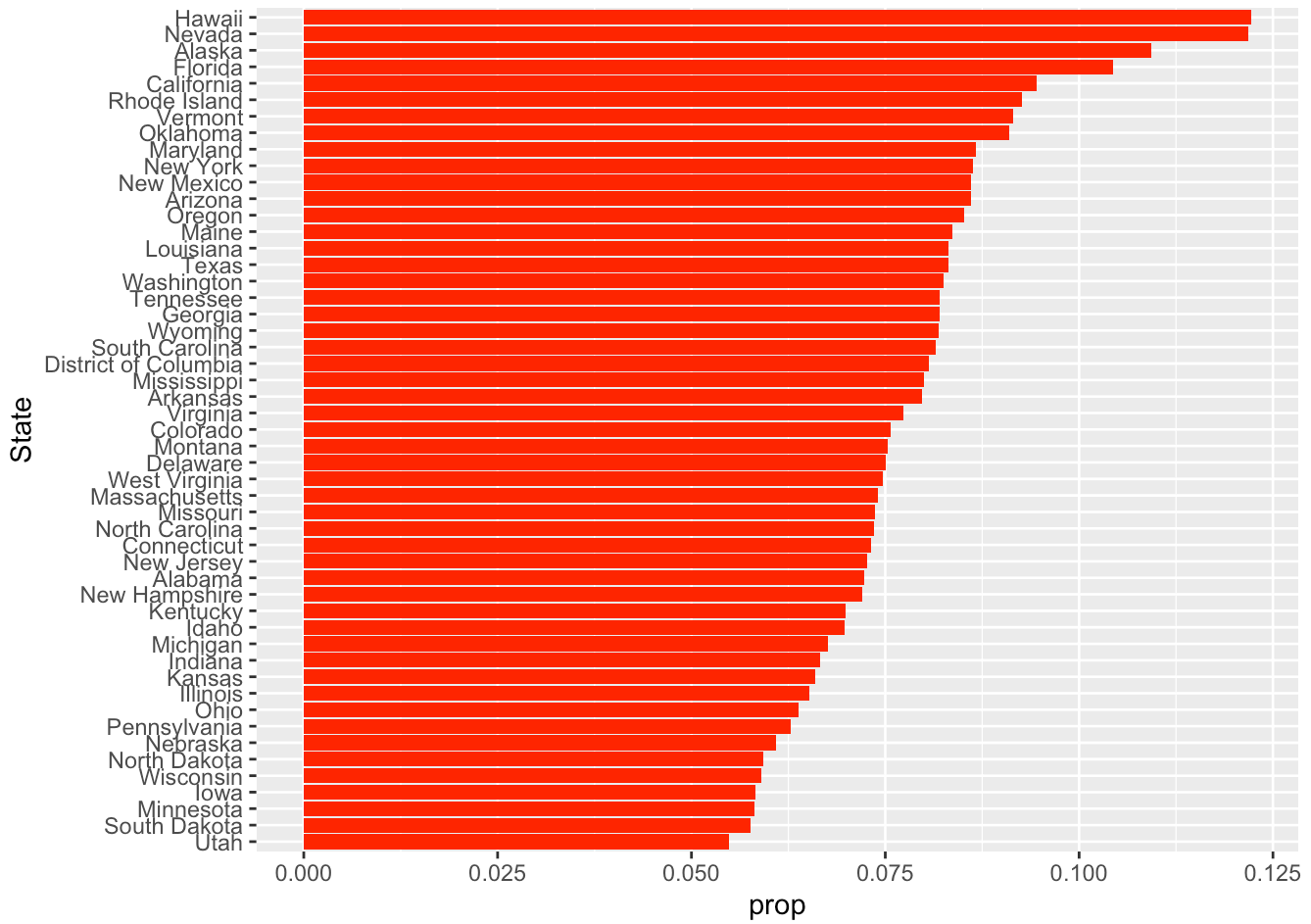
Proportion of Couples Where Man is 10+ Years Older By Men's Education



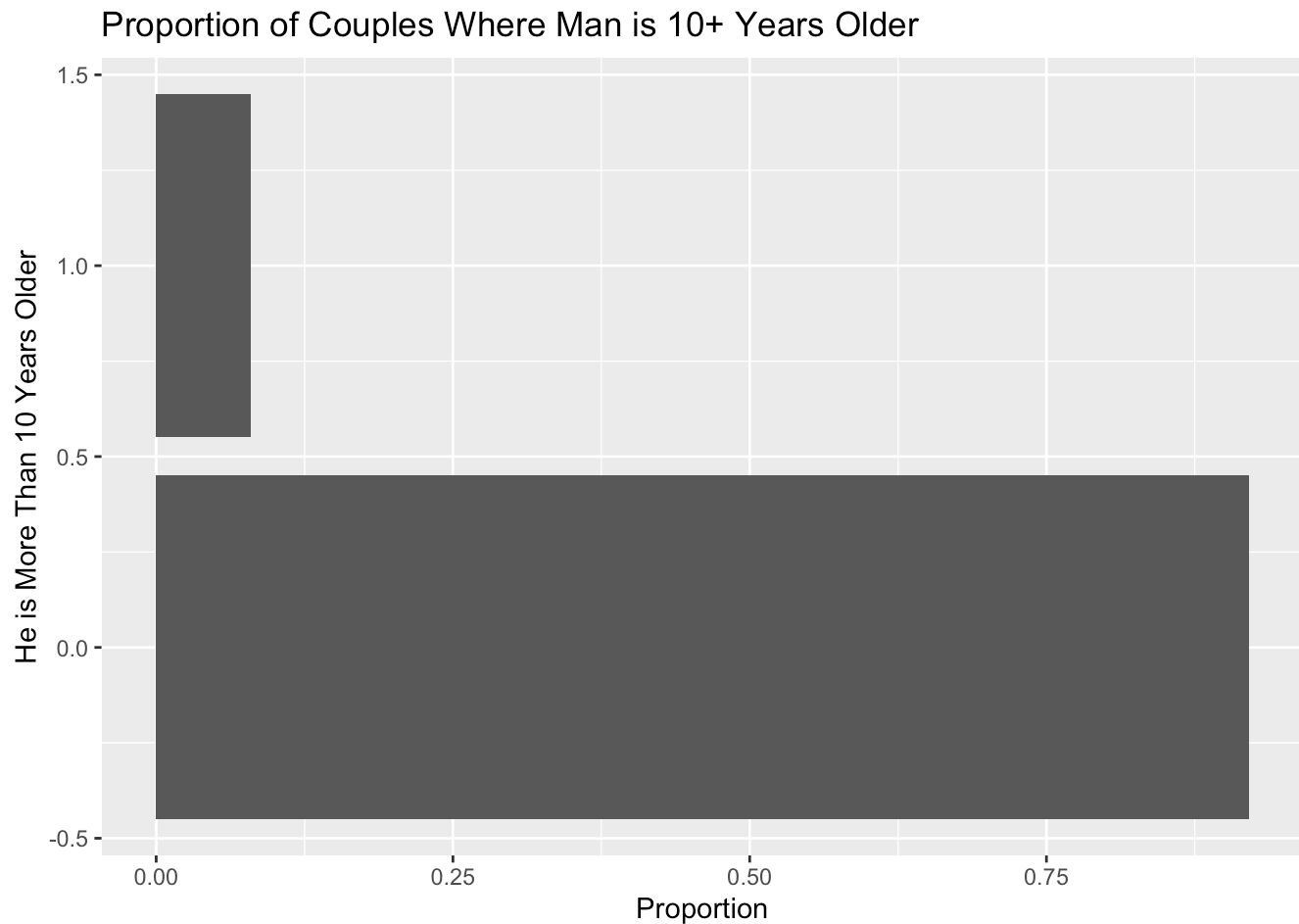
We also look at the proportion of those born in the US state against those not born in the US state. The graph indicates that a higher proportion of those not born in the US state have 10 years age gap between men and women. However, here the proportion for both are still very low.



We also see that different states may have an affect on the proportions of women married to men. Though, as we see with all the charts above, the overall proportion is still very low.



The overall low proportion trend we see in all the graphs above may be attributed to overall low percent of men who have married women 10 years or younger.



In our logistic model, we set a low threshold to account for a low volume of records where, in a traditional marriage, a woman's age is at least 10 years younger than the man. However, we notice that our model has a large bias towards predicting 0s. This may be due to an extreme low volume of records where a woman's age is at least 10 years younger than the man

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## threshold  
## 1 0.08011561
```

```
##
## Call:
## glm(formula = he_more_than_10yrs_than_her ~ same_race + RACE +
##       h_race + EDUC + STATEFIP, family = binomial, data = trad_data)
##
## Coefficients:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.824771    0.064081 -28.476 < 2e-16
## same_race        -0.380026    0.018079 -21.020 < 2e-16
## RACEBlack         0.413198    0.041159  10.039 < 2e-16
## RACEAmerican Indian or Alaska Native 0.116098    0.062569   1.856 0.063521
## RACEChinese       0.697289    0.055675  12.524 < 2e-16
## RACEJapanese     -0.012198    0.090543  -0.135 0.892837
## RACEOther Asian or Pacific Islander 0.839300    0.032758  25.621 < 2e-16
## RACEOther race    0.234575    0.037425   6.268 3.66e-10
## RACEtwo races     0.210466    0.023572   8.929 < 2e-16
## RACethree races   0.205267    0.080824   2.540 0.011095
## h_raceBlack       0.199137    0.039581   5.031 4.88e-07
## h_raceAmerican Indian or Alaska Native 0.047707    0.064169   0.743 0.457207
## h_raceChinese     -0.624120    0.066097  -9.443 < 2e-16
## h_raceJapanese    -0.337639    0.119450  -2.827 0.004704
## h_raceOther Asian or Pacific Islander -0.512169    0.037967 -13.490 < 2e-16
## h_raceOther race  -0.104633    0.039673  -2.637 0.008354
## h_racetwo races   -0.090637    0.023551  -3.849 0.000119
## h_racethree races -0.349797    0.088308  -3.961 7.46e-05
## EDUCNursery school to grade 4      0.294381    0.072115   4.082 4.46e-05
## EDUCGrade 5, 6, 7, or 8          -0.124466    0.050472  -2.466 0.013663
## EDUCGrade 9          -0.107385    0.065239  -1.646 0.099760
## EDUCGrade 10         -0.124283    0.061196  -2.031 0.042264
## EDUCGrade 11         -0.109382    0.059192  -1.848 0.064616
## EDUCGrade 12         -0.394270    0.036882 -10.690 < 2e-16
## EDUC1 year of college -0.466669    0.039461 -11.826 < 2e-16
## EDUC2 years of college -0.481604    0.040284 -11.955 < 2e-16
## EDUC4 years of college -0.737469    0.038044 -19.385 < 2e-16
## EDUC5+ years of college -0.858901    0.039525 -21.731 < 2e-16
## STATEFIPAlaska      0.403022    0.129868   3.103 0.001914
## STATEFIPArizona     0.176937    0.062147   2.847 0.004412
## STATEFIPArkansas    0.095350    0.078416   1.216 0.224004
## STATEFIPCalifornia   0.210989    0.052498   4.019 5.85e-05
## STATEFIPColorado    0.125403    0.065926   1.902 0.057148
## STATEFIPConnecticut 0.105209    0.075873   1.387 0.165551
## STATEFIPDelaware     0.068778    0.118620   0.580 0.562036
## STATEFIPDistrict of Columbia 0.192082    0.158375   1.213 0.225194
## STATEFIPFlorida     0.399524    0.053453   7.474 7.76e-14
## STATEFIPGeorgia     0.111458    0.059799   1.864 0.062339
## STATEFIPHawaii       0.444529    0.091576   4.854 1.21e-06
## STATEFIPIdaho       0.018960    0.094568   0.200 0.841094
## STATEFIPIllinois    -0.072646    0.058853  -1.234 0.217068
## STATEFIPIndiana     -0.040918    0.065382  -0.626 0.531433
## STATEFIPIowa        -0.147987    0.082102  -1.802 0.071470
## STATEFIPKansas      -0.034245    0.082370  -0.416 0.677594
## STATEFIPKentucky    0.034053    0.071514   0.476 0.633951
```

```

## STATEFIPLouisiana      0.105645    0.071469    1.478 0.139357
## STATEFIPMaine          0.272983    0.098202    2.780 0.005439
## STATEFIPMaryland       0.193058    0.064163    3.009 0.002622
## STATEFIPMassachusetts  0.131413    0.064399    2.041 0.041291
## STATEFIPMichigan      -0.008786    0.060563   -0.145 0.884650
## STATEFIPMinnesota     -0.128405    0.069337   -1.852 0.064040
## STATEFIPMississippi    0.062487    0.082094    0.761 0.446565
## STATEFIPMissouri      0.079033    0.065490    1.207 0.227512
## STATEFIPMontana       0.133862    0.111208    1.204 0.228704
## STATEFIPNebraska      -0.116582    0.097042   -1.201 0.229616
## STATEFIPNevada        0.454726    0.070691    6.433 1.25e-10
## STATEFIPNew Hampshire  0.125489    0.101903    1.231 0.218152
## STATEFIPNew Jersey    0.046147    0.060636    0.761 0.446621
## STATEFIPNew Mexico     0.149218    0.090649    1.646 0.099744
## STATEFIPNew York      0.230496    0.054766    4.209 2.57e-05
## STATEFIPNorth Carolina 0.040144    0.059441    0.675 0.499446
## STATEFIPNorth Dakota  -0.134495    0.139497   -0.964 0.334973
## STATEFIPOhio          -0.071264    0.059644   -1.195 0.232158
## STATEFIPOklahoma      0.183968    0.071929    2.558 0.010539
## STATEFIPOregon        0.215086    0.069349    3.101 0.001925
## STATEFIPPennsylvania  -0.070724    0.058634   -1.206 0.227744
## STATEFIPRhode Island   0.341445    0.107332    3.181 0.001467
## STATEFIPSouth Carolina 0.141063    0.067073    2.103 0.035453
## STATEFIPSouth Dakota  -0.150556    0.138281   -1.089 0.276255
## STATEFIPTennessee     0.170795    0.063116    2.706 0.006809
## STATEFIPTexas         0.113671    0.053592    2.121 0.033919
## STATEFIPUtah          -0.226586    0.083609   -2.710 0.006727
## STATEFIPVermont       0.406227    0.130623    3.110 0.001871
## STATEFIPVirginia      0.089346    0.060651    1.473 0.140722
## STATEFIPWashington    0.152169    0.061072    2.492 0.012716
## STATEFIPWest Virginia  0.103957    0.094092    1.105 0.269227
## STATEFIPWisconsin     -0.123390    0.067996   -1.815 0.069574
## STATEFIPWyoming       0.185812    0.142130    1.307 0.191096
##
## (Intercept)            ***
## same_race              ***
## RACEblack              ***
## RACEAmerican Indian or Alaska Native .
## RACEChinese            ***
## RACEJapanese
## RACEOther Asian or Pacific Islander ***
## RACEOther race         ***
## RACetwo races          ***
## RACethree races        *
## h_raceBlack            ***
## h_raceAmerican Indian or Alaska Native
## h_raceChinese          ***
## h_raceJapanese         **
## h_raceOther Asian or Pacific Islander ***
## h_raceOther race       **
## h_racetwo races        ***
## h_racethree races      ***

```



```

## EDUCNursery school to grade 4      ***
## EDUCGrade 5, 6, 7, or 8            *
## EDUCGrade 9                        .
## EDUCGrade 10                       *
## EDUCGrade 11                       .
## EDUCGrade 12                       ***
## EDUC1 year of college               ***
## EDUC2 years of college              ***
## EDUC4 years of college              ***
## EDUC5+ years of college             ***
## STATEFIPAlaska                     **
## STATEFIPArizona                     **
## STATEFIPArkansas                    .
## STATEFIPCalifornia                  ***
## STATEFIPColorado                   .
## STATEFIPConnecticut                 .
## STATEFIPDelaware                   .
## STATEFIPDistrict of Columbia        .
## STATEFIPFlorida                     ***
## STATEFIPGeorgia                     .
## STATEFIPHawaii                      ***
## STATEFIPIdaho                       .
## STATEFIPIllinois                    .
## STATEFIPIndiana                     .
## STATEFIPIowa                       .
## STATEFIPKansas                      .
## STATEFIPKentucky                    .
## STATEFIPLouisiana                   .
## STATEFIPMaine                       **
## STATEFIPMaryland                    **
## STATEFIPMassachusetts                *
## STATEFIPMichigan                    .
## STATEFIPMinnesota                   .
## STATEFIPMississippi                  .
## STATEFIPMissouri                    .
## STATEFIPMontana                     .
## STATEFIPNebraska                     .
## STATEFIPNevada                       ***
## STATEFIPNew Hampshire                 .
## STATEFIPNew Jersey                   .
## STATEFIPNew Mexico                   .
## STATEFIPNew York                     ***
## STATEFIPNorth Carolina                .
## STATEFIPNorth Dakota                  .
## STATEFIPOhio                         .
## STATEFIPOklahoma                     *
## STATEFIPOregon                       **
## STATEFIPPennsylvania                  .
## STATEFIPRhode Island                  .
## STATEFIPSouth Carolina                 .
## STATEFIPSouth Dakota                  .
## STATEFIPTennessee                    **

```

```
## STATEFIPTexas          *
## STATEFIPUtah           **
## STATEFIPVermont        **
## STATEFIPVirginia
## STATEFIPWashington     *
## STATEFIPWest Virginia
## STATEFIPWisconsin      .
## STATEFIPWyoming
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 229160  on 412274  degrees of freedom
## Residual deviance: 224194  on 412197  degrees of freedom
## AIC: 224350
##
## Number of Fisher Scoring iterations: 5
```

```
## Area under the curve: 0.6071
```

```
##
## pred      0      1
##    0 250679 16594
##    1 128757 16245
```